



Jahangirnagar University

Department of Computer Science and Engineering

2nd Year 1st Semester B.Sc. (Hons.) Final Examination -2024

Course Title: *Mathematics-III*

Course Code: *MATH-201*

Time: 3 Hours

Full Marks: 60

Answer any Six of the following questions

- Q1.** a) Define the gradient of a Scalar, the Divergence of a vector, and the curl of a vector. 5
If $\underline{a} = x^2z\hat{i} - 2y^3z^2\hat{j} + xy^2z\hat{k}$, find $\underline{\nabla} \cdot \underline{a}$ at the point (1,2,1).
- b) Prove that (i) $\underline{\nabla} \times (\underline{\nabla}\phi) = \underline{0}$, (ii) $\underline{\nabla} \cdot (\underline{\nabla} \times \underline{a}) = 0$. 5
- Q2.** a) Define directional derivative. Find the directional derivative of $\phi = x^2y^2z + 4xz^2$ at (1,2,1) in the direction $2\hat{i} + \hat{j} - 2\hat{k}$. 4
- b) Define solenoidal vector. Determine the constant a so that vector $\underline{v} = (x + 3y)\hat{i} + (y - 2z)\hat{j} + (x + az)\hat{k}$ is solenoidal. 3
- c) Show that $\underline{\nabla}\phi$ is a vector perpendicular to the surface $\phi(x, y, z) = c$ where c is a constant. 3
- Q3.** a) What is Argand diagram? Also, define modulus and argument of a complex number. 2
- b) What is analytic function? Find the necessary and sufficient condition for a complex function $f(z) = u(x, y) + iv(x, y)$ to be analytic at a point in a region R . 4
- c) Prove that $u = e^{-x}(x \sin y - y \cos y)$ is a harmonic function. Then find v such that $f(z) = u + iv$ is analytic. 4
- Q4.** a) State and prove the Cauchy's integral formula. 4
- b) Evaluate, using Cauchy's integral formula
$$\int_C \frac{e^{2z} dz}{(z-1)(z-2)}, \text{ where } C \text{ is the circle } |z| = 3. \quad 2$$
- c) State and prove Liouville's Theorem. 4

Q5. a) State Laurent Series.

2

b) Find the residues of $f(z) = e^z \operatorname{cosec}^2 z$ at all its poles in the finite plane.

3

c) Evaluate $\int_0^\infty \frac{dx}{x^4 + 81}$.

5

Q6. a) Find the Fourier Transform of $f(x) = e^{-|x|}$.

2

b) Define finite Fourier sine transform. Use finite Fourier transform to solve the boundary value problem

8

$$\frac{\partial U}{\partial t} = 3 \frac{\partial^2 U}{\partial x^2}$$

$$U(0, t) = 0, U(2, t) = 0, t > 0$$

$$U(x, 0) = x, 0 < x < 2.$$

Q7. a) Define Fourier integral, Fourier Cosine and Sine integral.

3

b) Find the Fourier integral of

7

$$f(x) = \begin{cases} |x|; & -\pi \leq x \leq \pi \\ 0; & |x| > \pi \end{cases}$$

Q8. a) Define Laplace transform. Find the Laplace transform of the function $F(t) = t^n e^{at}$.

4

b) Solve the following differential equations by using Laplace transform

6

i. $\frac{dx}{dt} + 5x = e^{2t}, \quad x(0) = 1$

ii. $\frac{d^2 y}{dt^2} + y = 3 \sin(2t), \quad y(0) = 3, y'(0) = 1.$



Jahangirnagar University
Department of Computer Science and Engineering
2nd Year 1st Semester B.Sc. (Hons.) Final Examination -2024

Course Title: **Computer Ethics and Cyber Law**
Time: **3 Hours**

Course No: **CSE-203**
Full Marks: **60**

[Answer each of the following questions. Each question carries equal marks. Figures in the right margin indicate marks.]

1.

Section-A (CO1)

(Answer *all* of the following questions.)

- a) **Define** the following terms related to intellectual property: (i) Competitive intelligence (ii) Reverse engineering. 2
- b) **Define** the following terms relating to privacy and security: (i) Microtargeting (ii) Identity theft (iii) Spear-phishing (iv) Spyware 2
- c) **What** are viruses? **Explain** how an email virus spreads. 2
- d) **Define** *digital divide* and *net neutrality*? Why is it important to bridge the digital divide? 2
- e) **What** is the digital divide? Where does it exist? **Why** is it important to bridge the digital divide? 2
- f) **When** does zero attack take place? 2

2.

Section-B (CO2)

(Answer any *four* of the following questions)

- a) What is open source software movement? **Explain** its advantages and limitations. 3
- b) How can a well-implemented program can reach the goal of reducing ethics risk? **Explain** with a flowchart. 3
- c) Identify and **explain** three common ethical issues for IT users. And also **suggest** three actions that can be taken by companies to support ethical practices of IT users. 3
- d) **Why** computer related security incidents are so prevalent now-a-days? **Identify** few causes and **explain**. 3
- e) Answer the followings: 3
 - (i) Identify few advantages and limitations of telework / work-from-home from the worker's point-of-view [1 mark]
 - (ii) Discuss the dilemma between moral responsibility and personal/organizational risks of whistle-blowing. [2 marks]

3.

Section-C (CO3)

(Answer any *three* of the following questions)

- a) **Compare** among the following terms: (i) Morals (ii) Ethics (iii) Law (iv) Virtue (v) vice 4
- b) **Describe** similarities and differences between social contract theory and act utilitarianism. 4
- c) **Describe** the strengths and limitations of using copyrights, patents, and trade secret laws to protect intellectual property? 4
- d) **Compare** among different classes of perpetrators of computer crime. 4

4.

Section-D (CO4)

(Answer any *three* of the following questions)

- a) *"Dhaka University, Jahangirnagar University, and Rajshahi University denied admission to more than 100 PhD applicants because they took an online peek at the status of their applications. These students learned how to circumvent the program's security, and they used this knowledge to view their files and see if they had been accepted. Students could see information about their own application, but could not view the status of other students' applications. In many cases the students learned that no admission decision had yet been made."* 4

Do you feel the response of these universities was appropriate? Use the five-step decision-making process to **analyze** the above situation and **recommend** a course of action. Which option would you pursue and why? And why do you not **choose** the other options?

- b) *"Students in a Computer Programming class are asked to take a quiz posted on the google classroom. The instructor has explained the following rules to the students: First, they are supposed to do their own work. Second, they are free to consult their lecture notes and the textbook while* 4

taking the quiz. Third, in order to get credit for the quiz, they must correctly answer at least 80 percent of the questions. If they do not get a score of 80 percent, they may retake the quiz as many times as they wish.

Ridita and Zakir are both taking the quiz. They are sitting next to each other in the computer room. Zakir asks Ridita for help in answering one of the questions. He says, "What's the difference if you tell me the answer, I look it up in the book, or I find out from the computer that my answer is wrong and retake the quiz? In any case, I'll end up getting credit for the right answer." Ridita tells Zakir the correct answer to the question."

Do you think Ridita did the right thing? **Analyze** and **justify** the morality of Ridita's decision based on "Kantianism" ethical theory.

- c) "Rifat and his friends, aged between 13 and 17, were very fond of action-oriented, role-playing computer games. After some time, severe changes were noticed in their behavior. They wanted to play the games for longer durations. They became very aggressive, started using offensive language, and often got into conflicts with their peers on minor issues. A psychiatrist suggested that the violence that the children observed in the computer games had a negative effect on them." 4

Do you think it is ethically correct for a programmer or a company to produce such games? Consider the software engineering code of ethics to **analyze** the situation and **recommend** a course of action.

- d) **Explain** the ethical relationship between IT professionals and suppliers. Consider the following Scenario: 4

"You are in charge of awarding all computer hardware service contracts (valued at over \$2 million per year) for your employer. In recent emails with the company's current service contractor, you casually exchanged ideas about family vacations. You mentioned that your family is planning on vacationing in the Sreemangal area. You are surprised when the contractor emails you an offer to use his company's condominium at a Sreemangal resort, complete with golf and health club privileges. He assures you that the condo would normally be empty that time of year and that other customers frequently use the condo. The resort is one you are familiar with but have never used because the rental is well over \$5,000 per week. You would really like for your family to experience staying at a five-star resort but you worry about the potential consequences of accepting the offer."

Could this offer be considered a bribe? What would you do? **Analyze** the situation considering the software engineering code of ethics.

5.

Section-E (CO5)

(Answer any two of the following questions)

- a) Consider a cybercrime scenario as below:

"Faria is a second-year CSE student at Netrokona University. She has disability on her left hand which is weak from her birth. Her family is not solvent enough to support her education, so she has to look for an alternate income source. As a computer science student, she came up with the idea of making a mobile application for tuition seekers. Many university students and guardians made accounts in her app. Later a shop came to visit her and offered her a handsome amount of money to share app users' personal information so they could use it for personalized business advertisements. Faria badly needed the money and could not refuse the offer." 6

Considering the above scenario, answer the followings:

- (i) Do you think Faria did something wrong? **Judge** your opinion in using one or more appropriate ethical theory. [3 marks]
- (ii) According to "Cyber Security Act-2023", analyze the above scenario and identify the rules that have been broken (if any) by Ms. Faria and the corresponding punishments she will be subject to. [3 marks]

- b) *"In 2010 Albert Gonzalez was sentenced to 20 years of imprisonment after pleading guilty to using an SQL injection attack to steal more than 130 million credit and debit card numbers. Some of the credit and debit card numbers were sold online, leading to unauthorized purchases and bank withdrawals. The targets of the attacks were Heartland Payment Systems, 7-Eleven, Hannaford Brothers Supermarkets, TJX, DSW, Barnes & Noble, OfficeMax, and the Dave & Buster's chain of restaurants. Most of the numbers were stolen from Heartland Payment Systems, which estimated its losses at \$130 million."* 6

Answer the followings:

- i) Following the above story, **create** a real-like cybercrime scenario in the context of Bangladesh relating identity theft. [2 marks]
- ii) **Judge** the scenario in (i) considering one appropriate ethical theory. [2 marks]
- iii) Do you think the current Digital Security Act in Bangladesh will cover the cybercrime in (i)? If so, **identify** the clauses are in consideration for this crime and the corresponding offences and punishments. [2 marks]

c) Answer the followings: 6

- (i) Consider the software code of ethics to **analyze** the following situation and **recommend** a course of action. [4 marks]
- (ii) Also **judge** the situation using Richard DeGeorge's questions for whistle-blowing. [2 marks]

"Company X wants to open a dating service Web site. It hires Company Y to develop the software. Company Y hires Gina as a private contractor to provide a piece of instant messaging software for the package. Gina's contract says she is not responsible for the security of the site. Company Y is supposed to perform that bit of programming. However, software development runs behind schedule, and Company Y implements a simplistic security scheme that allows all messages to be sent in plain text, which is clearly insecure."

Gina brings her concerns to the management of Company Y. Company Y thanks her for her concern, but indicates it still plans to deliver the software without telling Company X. Company Y reminds Gina that she has signed a confidentiality agreement that forbids her from talking about the software to anyone, including Company X."

----- End of question script -----



Jahangirnagar University
Department of Computer Science and Engineering
2nd Year 1st Semester B.Sc. (Hons.) Final Examination -2024

Course Title: *Numerical Methods*
Time: *3 Hours*

Course No: *CSE-205*
Full Marks: *60*

[Answer each of the following questions. Figures in the right margin indicate marks.]

1.

Section-A (CO1)

(Answer *all* of the following questions.)

- a) **Identify** situations in numerical computation where instability may occur. 2
- b) **Define:** i) Accuracy ii) Precision 2
- c) **Name and define** two sources of numerical errors in computations. 2
- d) **Define** curve fitting. 2
- e) **Define** interpolation. What are the methods available for interpolation? 2
- f) **Compare** critically Gauss elimination and Gauss-Jordan methods of solving simultaneous equations. 2

2.

Section-B (CO2)

(Answer any *three* of the following questions)

- a) An electrical engineer is designing a circuit with a nonlinear resistance function. The resistance $R(x)$ of the circuit depends on the voltage x applied across it, and is given by the equation: 4
$$R(x) = x^3 - 6x^2 + 9x - 4$$

The engineer needs to determine the voltage x at which the resistance $R(x)$ equals zero, which would indicate a malfunction in the circuit. The voltage x must be found within the range ~~$1 \leq x \leq 4$~~ $2 \leq x \leq 5$. Using the Bisection Method, find the voltage x where the resistance $R(x)=0$. **Illustrate** at least four iterations to determine the root, accurate to four decimal places.
- b) A computer engineer is working on a robotics system where the robot must adjust its arm position in real-time using inverse kinematics. The nonlinear equations involved do not have closed-form solutions, so the engineer must use numerical methods to solve them efficiently. **Give** a block diagram explaining the process from equation formulation to solution in the robot's control system. 4
- c) **Differentiate** between interpolation and regression with proper examples. 4
- d) An electrical engineer is working on optimizing power distribution in a city's power grid. The system of linear equations representing the power flow is large. 4
 - The Jacobi method is initially chosen because it allows parallel computation of updates for different nodes.
 - However, it is observed that the method takes many iterations to converge.
 - The engineer considers switching to the Gauss-Seidel method to speed up convergence.

Identify the appropriate method should the engineer use for faster results, and also **give** the proper reasons for that.

3.

Section-C (CO3)

(Answer any *three* of the following questions)

- a) **Classify** the four possible solution conditions of a system of linear equations but it may not be unique. Explain each one of them with proper illustration. 4
- b) **Solve** the given system of equations using Matrix-Inversion method. 4
$$\begin{aligned} 2x + 4y - 6z &= -8 \\ x + 3y + z &= 10 \\ 2x - 4y - 2z &= -12 \end{aligned}$$
- c) A game developer is working on a physics-based simulation where an object's movement requires computing the square root of non-integer values for speed calculations. The game's performance depends on quick approximations instead of expensive square root computations. The developer has a lookup table for square roots of integers: 4

x	1	2	3	4	5
$\sqrt{x}$	1	1.4142	1.7321	2	2.2361

To **calculate** the square root of 2.5 for the game's physics engine, they decide to use **second order Lagrange interpolation polynomial**. Now **compute** the estimated value.

- d) **Derive** a formula for computing second derivative of a function using Tylor's expansion. Find approximation to second derivative of $\cos(x)$ at $x=0.75$ with $h=0.01$. Compare with the true value. 4

4.

Section-D (CO4)

(Answer any *three* of the following questions)

- a) A small manufacturing company is analyzing its production output to understand how increasing the number of machines affects the total units produced. They have collected the following data: 4

Number of Machines (x)	1	2	3	4	5
Units Produced (y)	0.5	2	4.5	8	12.5

Assuming the relationship between the number of machines and units produced follows a power function of the form $y = ax^b$, **calculate** the values of constants a and b that best fit the data.

- b) A drone navigation system relies on accurate estimations of angular velocities to adjust its orientation. The onboard computer estimates the rate of change of the angle θ , where $\theta(t) = \sin(t)$, with t in radians, at a particular moment when $t=0.45$ seconds. 4

To compute the numerical derivative at increasing values of h from 0.01 to 0.04 with a step size of 0.005. **Analyze** the variation of error in each step.

- c) An engineer is designing a sensor that measures the exponential growth of voltage across a capacitor in a circuit, where the voltage at time x seconds is modeled by the function $v(x)=e^x$. To estimate the total voltage accumulated over a time interval from $x = -1$ to $x = +1$ seconds, the engineer decides to use the **Composite Trapezoidal Rule** due to its computational simplicity. 4

Solve the following approximate integral using:

$$\int_{-1}^1 e^x dx$$

- (i) $n=2$ equally spaced time intervals.
(ii) $n=4$ equally spaced time intervals.
- d) A space research team is simulating the motion of a satellite re-entering Earth's atmosphere. The total heat generated on the satellite's surface is modeled by a complex function involving altitude, velocity, and varying atmospheric density: 4

$$Q(t) = \int_0^T f(t, v(t), h(t)) dt$$

- (i) **Analyze** why traditional methods of calculus (analytical integration) are not feasible for computing the total heat $Q(t)$ in this scenario.
(ii) **Figure out** at least one advantage and one trade-off of using numerical integration in real world engineering problems.

5.

Section-E (CO5)

(Answer any *two* of the following questions)

- a) For the following differential equation, **estimate** $y(2)$ using i) Euler's method and ii) Heun's method using $h = 0.25$. Compare the results with exact answers. 6

$$y'(x) = \frac{2y}{x} \quad \text{with } y(1) = 2.$$

- b) An automotive engineer is analyzing the performance of a prototype electric car during a controlled acceleration test. The car's position was recorded at one-second intervals, yielding the following data: 6

Time, $t(s)$	5	6	7	8	9
Distance, $s(t)$ (km)	10.0	14.5	19.5	25.5	32.0

Estimate the instantaneous velocity of the car at $t = 5$, $t = 7$, and $t = 9$ seconds using appropriate finite difference methods:

- (i) Use the forward difference method for $t=5$.
(ii) Use the central difference method for $t=7$.
(iii) Use the backward difference method for $t=9$.
- c) The sales in a particular shop for various years is given in the following table. 6

Year:	2010	2012	2014	2016	2018
Sales (in lakhs)	40	43	48	52	57

Estimate the sales for the year 2015 using i) Newton's forward difference interpolation formula and ii) Newton's backward difference interpolation formula.

— End of question script —



Jahangirnagar University
Department of Computer Science and Engineering
2nd Year 1st Semester B.Sc. (Hons.) Final Examination -2024

Course Title: *Algorithms-I*

Time: 3 Hours

Course Code: CSE-209

Full Marks: 60

[Answer each of the following sections. Each section carries equal marks. Figures in the right margin indicate marks.]

Section-A (CO1)

Answer each of the following questions.

- 1 a) **Define** algorithm. State its key characteristics. 2
- b) **Show** that, any decision tree that can sort n elements must have $\Omega(n \lg n)$ comparisons. 2
- c) Recall the properties of greedy algorithm. 2
- d) **Define** each of the following with example. 2
 - i. Stable Sort
 - ii. In-place Sort
- e) You have the following set of coins: [1, 2, 5, 10, 20]. Now write an algorithm to **find** out a way to give change an amount to a customer so that, total number of coins minimized. 2
- f) Compare between quick sort and randomized quick sort along with their complexity and limitations. 2

Section-B (CO2)

Answer any **three** of the following **four** questions.

- 2 a) **Outline** an algorithm to merge arrays $a[m]$ (sorted in *ascending* order) and $b[n]$ (sorted in *descending* order) into third array $C[n + m]$ in ascending order. 4
- b) **Demonstrate** the following recurrence relations using Master method. 4
 - i. $T(N) = 2T\left(\frac{N}{4}\right) + \sqrt{N}$
 - ii. $T(N) = 4T\left(\frac{N}{2}\right) + N^2 \lg N$
- c) You are given the following adjacency list representation of an undirected graph. 4

Now, **illustrate** the graph traversing using Breadth-First order, starting from vertex C. Whenever you examine neighbors of a vertex, process the nodes according to the adjacency list (from the smallest to largest vertex number). Fill in the table (the vertices, edges and the distance).

A: B, C, D, H

B: A, C, E, H

C: A, B, G

D: A, F

E: B, G

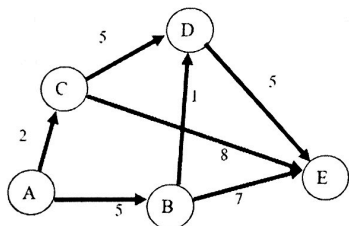
F: D

G: C, E

H: A, B

Order from first to last edges added to the BFS tree	Vertex, v, (in order visited by BFS)	Edge: u, v (edge that adds v to the BFS tree)	Distance (as number of edges) from the source
1 st	C		
...			

- d) Consider the following graph and answer the questions:



Graph-1

- i. If possible, **list** two valid topological orderings of the nodes in the graph above. If there is only one valid topological ordering, list that one ordering. If there is no valid topological ordering, state why one does not exist.

- ii. Which of the following are correct about the graph.

a. complete	b. directed	c. strongly connected
d. weighted	e. acyclic	f. weakly connected

Section-C (CO3)

Answer any **three** of the following **four** questions.

- 3 a) Consider the following array elements.

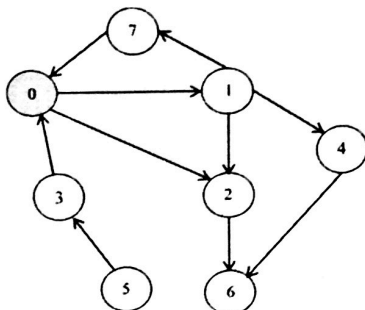
Data	80	60	40	8	65	22	28	74	6	12	2	13
Index	1	2	3	4	5	6	7	8	9	10	11	12

Construct the Heap from the array, does the following array represent a **Max-Heap** with root at index 1? If yes, explain why it is a Max-Heap. If not, at which indices should we call the **Max-Heapify** function? How many times should we **Max-Heapify** to make it a Max Heap.

b) **Solve** Rock Climbing Problem for the following data set.

2	4	5	1	7
4	5	3	8	8
1	3	6	3	9
15	8	6	12	7

c)



Graph-2

Consider the left graph and answer the following:

- i. Traverse this graph by **applying** Depth First Search (DFS) starting from node 0. If a vertex has more than one neighbor you must visit them in an increasing order of vertex number, for example from node 1 you would visit the neighbors in order: 2,4,7.

Order from first to last	Visited vertex	Parent
1 st	0	-
2 nd		

- ii. Label the edges as one of these categories: T (tree), B (backward), C (cross) or F(forward).

d) **Identify** the longest increasing subsequence from the given sequences.

$$S = \{10, 22, 9, 33, 21, 50, 41, 60, 80\}$$

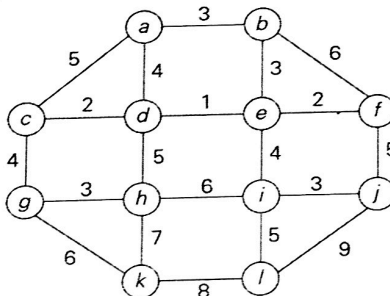
Section-D (CO4)

Answer any **three** of the following **four** questions.

- 4 a) Analyze the application for Huffman coding in the field of computer science within two sentences. Hence, point out the length of the codeword using Huffman coding.

	a	b	c	D	e	f
Frequency (thousands)	34	14	12	16	14	10

- b) Apply Dijkstra, the single source shortest path algorithm on Graph-3 from source **a**, analyze the shortest path construction process.



Graph-3

- c) Find Minimum Spanning Tree for above Graph-3 using Kruskal algorithms
- d) You are given two strings, X="AGGTAB" and Y="GXTXAYB". Now, calculate the LCS between them using DP table.

Section-E (CO5)

Answer any **two** of the following **three** questions.

- 5 a) **Bhombol** is an adventurer, today he is preparing for a journey across a treacherous mountain. He has a backpack that can carry at most 10 kg of supplies. In front of him lies a collection of 6 different items, each with its own weight and usefulness value. He knows he can only choose each item once. His goal is simple, maximize the total usefulness of what he carries, without exceeding the 10 kg limit of his knapsack. Now, **evaluate** the optimal solution for **Bhombol**.

Item	Usefulness Value	Weight
1	20	1
2	5	2
3	10	3
4	40	8

5	15	7
6	25	4

- b) Consider the following 15 edges of a directed weighted graph, where each edge has (source, destination, cost) format. 6

(A, B, 2)	(A, F, -6)	(B, C, 20)	(B, G, 1)	(C, H, 5)	(D, C, 4)	(D, H, 6)	(E, A, 20)
-	(E, F, 10)	(F, A, 7)	(F, B, 3)	(F, G, 11)	(G, C, 9)	(G, D, 4)	(G, H, 17)

Now, **determine** the shortest path from vertex E to vertex H using the Bellman-Ford algorithm based on the above lexicographical edge order. To do so, complete the following table where It represents iteration, d[n] represents minimum distance to n from E and p[n] represent the parent of n.

	Initial		It-1		It-2		It-3		It-4		It-5		It-6		It-7	
	d[n]	p[n]	d[n]	p[n]	d[n]	p[n]	d[n]	p[n]	d[n]	p[n]	d[n]	p[n]	d[n]	p[n]	d[n]	p[n]
A	∞	0														
B	∞	0														
C	∞	0														
D	∞	0														
E	0	-1														
F	∞	0														
G	∞	0														
H	∞	0														

- c) Trace the sequence of operations below, using a disjoint set with union by size (*without path compression*). If there is a tie when performing a union, the tree with the smaller-valued root should be the root of the union. Assume there are initially 10 sets {1},{2},{3},...,{10}. 6

```

union(7, 8);
union(3, 6);
union(2, 5);
union(1, 10);
union(1, 4);
union(3, 7);
union(2, 7);
union(10, 9);
x = find(7);
y = find(9);
union(x, y);

```

- Interpret** the final forest of up-trees that result from the operations above.
- Interpret** the new forest of up-trees that results from doing a find(9) *with path compression* on your forest of up-trees from (a).



Jahangirnagar University
Department of Computer Science and Engineering
2nd Year 1st Semester B.Sc. (Hons.) Final Examination -2024

Course Title: **Electronic devices and circuits-II**
Time: 3 hours

Course No: **CSE-207**
Full Marks: **60**

[Answer each of the following questions. Each question carries equal marks. Figures in the right margin indicate marks.]

1.

Section-I (CO1)

Answer all the questions.

- | | |
|---|---|
| a) $CMRR (\log) = 20 \log_{10} A_d/A_c$ is an expression for what purpose? Elaborate on each term. | 2 |
| b) State Programmable Unijunction Transistor (PUT), Opto-Isolators, Varactor (Varicap) Diodes, and Triac application areas. | 2 |
| c) List the basic features of Op-Amp. | 2 |
| d) Give the concept of Quasi-Complementary Push Complementary Push-Pull Amplifier. | 2 |
| e) Sketch the terminal identification of SCR. | 2 |
| f) Draw graphic symbols for (i) Schottky Diode and (ii) Tunnel Diodes. Also, state at least one use of each diode. | 2 |

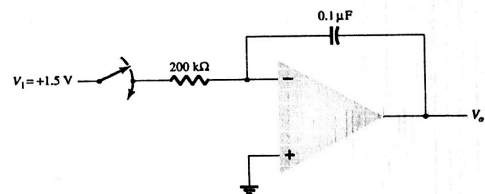
2.

Section-II (CO2)

Answer any three of the following questions.

a)

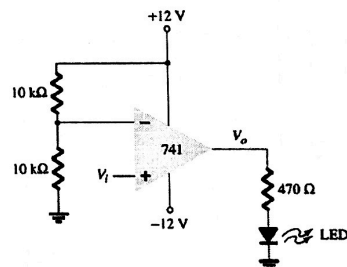
- i. What type of operation above op-amp circuit perform?
- ii. State the equation for the output voltage of the circuit.
- iii. Sketch the output waveform resulting in Figure.



4

- b) The circuit is a non-inverting comparator circuit application.

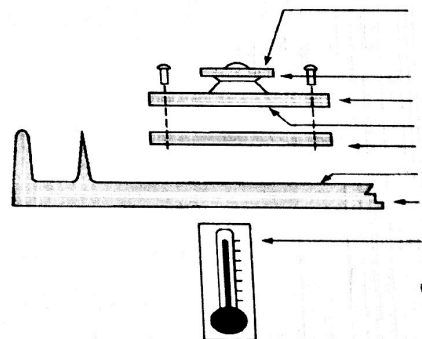
- i. Describing the features non-inverting comparator.
- ii. State the operation of the circuit, i.e., when the LED turns on or off.



4

- c) i.

More details about the thermal characteristics of the transistor and its relation to the power dissipation of the transistor may help provide a clearer understanding of power as limited by temperature. The bottom is the ambient temperature, and the top is the source of heat. Now label all the items and the relevant temperatures. Also, identify an equivalent thermal to electrical analogy.

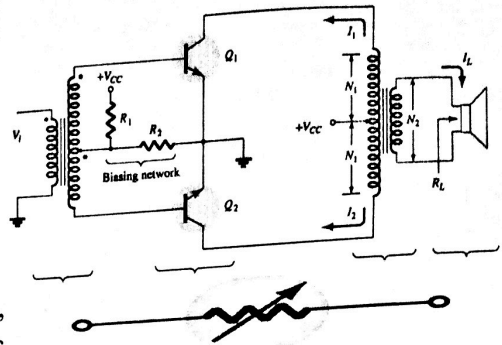


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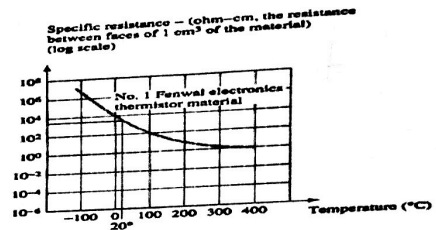
OR

ii)

What type of amplifier is the above circuit?
The biasing circuit is identified. Now identify the sections of the circuit. Clearly state the purpose of all sections.



- d) What type of device is it? For this device, determine the dynamic rate of change in specific resistance with temperature at $T = 20^\circ\text{C}$. How does this compare to the value determined at $T = 300^\circ\text{C}$? From the results, determine whether the greatest change in resistance per unit change in temperature occurs at lower or higher levels of temperature. Note the vertical log scale.

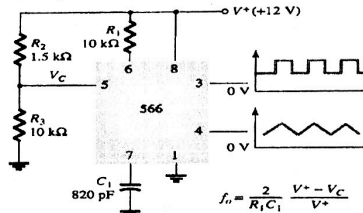


3.

Section-III (CO3)

Answer any three of the following questions.

- a) Show the annotated circuit diagram of three op-amp stages using an LM348 IC to provide 4 outputs that are 10, 20, and 50 times larger than the input. Use a feedback resistor $R_f = 500\text{ k}\Omega$ in all stages.
- b) After explaining the operation of the voltage-controlled oscillator as shown in the following figure, find the frequency of the square wave.



- c) A pure sinusoidal signal has a single frequency at which the voltage varies positive and negative by equal amounts. Any signal varying over less than the full 360° cycle is considered to have distortion.
- What are the main causes of distortion?
 - Calculate the harmonic distortion components for an output signal having a fundamental amplitude of 2.5 V, second harmonic amplitude of 0.25 V, third harmonic amplitude of 0.1 V, and fourth harmonic amplitude of 0.05 V.
 - Calculate the total harmonic distortion for the amplitude components given above.
- d) i. Sketch a five-stage ladder network using 15-k Ω and 30-k Ω resistors.
- ii. For a reference voltage of 16 V, calculate the output voltage for an input of 11010 to the circuit of Problem i.

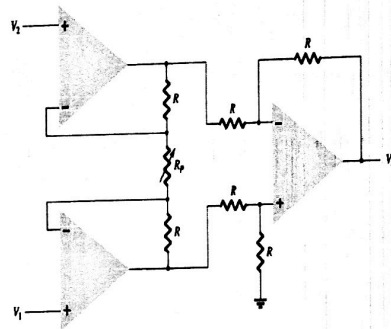
Section-IV (CO4)

Answer any three of the following questions.

4. a) Feedback amplifier types: (i) voltage-series feedback, $A_f = \underline{\hspace{1cm}}$ (ii) voltage-shunt feedback, $A_f = \underline{\hspace{1cm}}$; (iii) Current-series feedback, $A_f = \underline{\hspace{1cm}}$, and (iv) Current-shunt feedback, $A_f = \underline{\hspace{1cm}}$. State the effect of Feedback Connection on Input and Output Impedance for different types of feedback amplifiers.

- b) The circuit providing an output based on the difference between two inputs (times a scale factor) is shown in the Figure. A potentiometer is provided to permit adjusting the scale factor of the circuit. Whereas three op-amps are used, a single-quad op-amp IC is all that is necessary (other than the resistor components). The output voltage can be expressed as the equation shown above.

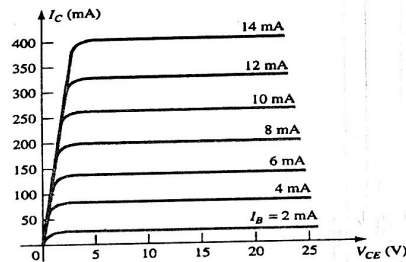
$$V_o = \left(1 + \frac{2R}{R_p}\right)(V_1 - V_2) = k(V_1 - V_2)$$



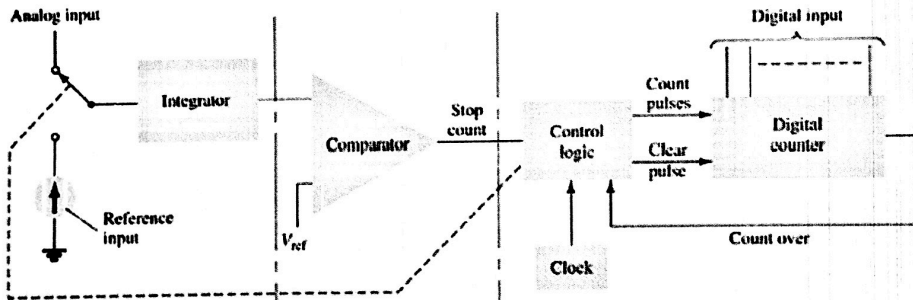
- Explain the operation of this instrument amplifier.
- Calculate the output voltage for the circuit if $R = 5K$ and $R_p = 500 \text{ Ohm}$; $v_1 = 50\text{mv}$ and $v_2 = 12\text{mv}$

- c) The above figure shows the Transistor characteristic.

- After redrawing the characteristics curve, place the Q-point for class A, class B, class C, and class AB power amplifiers over a load line.
- Which one has high gain? What is a drawback of this type of amplifier?



d)



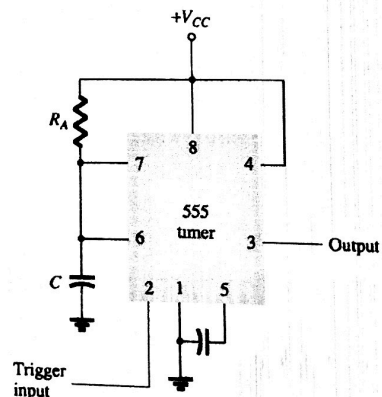
This is an Analog-to-digital conversion using a dual-slope method logic diagram. Clearly explain operation of each block. Also show example of small voltage and large voltage conversion.

Section-V (CO5)

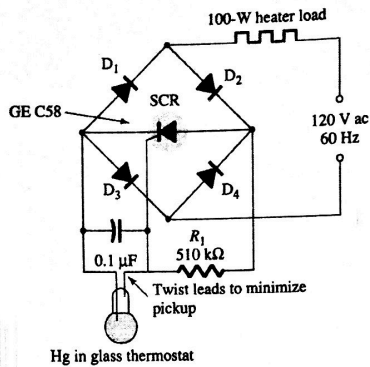
Answer any two of the following questions.

5. a) The 555 timer can also be used as a one-shot or monostable multivibrator circuit, as shown in the figure.

- Explain the operation of the circuit.
- Draw the output waveforms with the trigger input.
- Determine the period of the output waveform for the circuit when triggered by a negative pulse.
- Convert this circuit so that it operates in an astable mode.



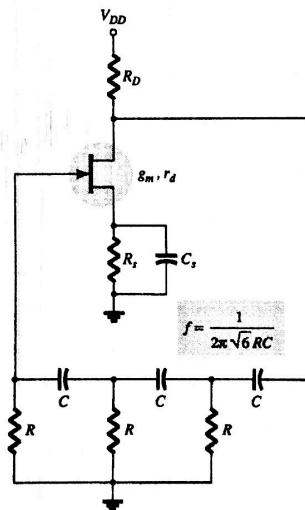
b)



A schematic diagram of a 100-W heater control using an SCR appears in Figure above. It is designed such that the 100-W heater will turn on and off as determined by thermostats.

- Mercury-in-glass thermostats are very sensitive to temperature change. They can sense changes as small as 0.1°C .
- Explain the operation of the circuit
- Sketch the waveform of the full-wave rectified waveform across the SCR.
- Once the thermostat reaches its set temperature and assumes the short-circuit state, how will the SCR react?
- Is it possible to use TRIAC instead of SCR? What is your comment?

c)



- This is a FET phase-shift oscillator. An FET phase-shift oscillator having $g_m = 6000 \mu\text{S}$, $r_d = 36 \text{ k}\Omega$, and feedback resistor $R = 12 \text{ k}\Omega$ is to operate at 2.5 kHz . Design the C for the specified oscillator operation.
- From FET amplifier theory, the amplifier gain magnitude is calculated from $|A| = g_m R_L$ where R_L in this case is the parallel resistance of R_D and r_d . Find R_D for $A > 29$ to ensure oscillator action.
- An op-amp can be used in a crystal oscillator. The crystal is connected in the series-resonant path and operates at the crystal series-resonant frequency. Can you design such a circuit?

Rohan

What is the time complexity of Binary Search in the worst case?

- A. $O(\log n)$ B. $O(n)$
C. $O(n \log n)$ D. $O(n^2)$

Which sorting algorithm has the best average case time complexity?

- A. Bubble Sort B. Insertion Sort
C. Quick Sort D. Selection Sort

Which of the following algorithms is used to find the shortest path in a graph?

- A. Prim's Algorithm B. Dijkstra's Algorithm
C. Kruskal's Algorithm D. Bellman-Ford Algorithm

In which scenario is the Fractional Knapsack problem more optimal than the 0/1 Knapsack problem?

- A. When items are divisible
B. When items are indivisible
C. When there are no constraints on item weights
D. When profit values are integers

Which of the following algorithms works by dividing a problem into smaller sub-problems and then solving them recursively?

- A. Dynamic Programming B. Greedy Algorithm
C. Divide and Conquer D. Brute Force

What is the time complexity of Merge Sort in the worst case?

- A. $O(n)$ B. $O(n \log n)$ C. $O(\log n)$ D. $O(n^2)$

Which of the following algorithms uses a priority queue to efficiently find the shortest path in a graph?

- A. Kruskal's Algorithm B. Prim's Algorithm
C. Dijkstra's Algorithm D. Bellman-Ford Algorithm

Which algorithm can be used to find the longest increasing subsequence in an array?

- A. Binary Search B. Dynamic Programming
C. Greedy Algorithm D. Backtracking

In Kruskal's Algorithm, which data structure is used to detect cycles in a graph?

- A. Stack B. Queue C. Disjoint Set D. Priority Queue

Which algorithm is commonly used to perform a breadth-first search (BFS) on a graph?

- A. Stack B. Queue C. Priority Queue D. Heap

What is the space complexity of the Depth First Search (DFS) algorithm?

- A. $O(V)$ B. $O(V + E)$ C. $O(V^2)$ D. $O(E)$

What is the key difference between Prim's and Kruskal's algorithm?

- A. Kruskal's Algorithm selects the minimum edge, while Prim's Algorithm selects the minimum vertex
B. Kruskal's Algorithm works only for directed graphs
C. Prim's Algorithm works only for weighted graphs

D. Prim's Algorithm uses a priority queue, while Kruskal's Algorithm does not

Answer: A

Which of the following algorithms is used to compute the transitive closure of a graph?

- A. Dijkstra's Algorithm
B. Floyd-Warshall Algorithm
C. Prim's Algorithm
D. Kruskal's Algorithm

Which of the following algorithms can detect negative weight cycles in a graph?

- A. Dijkstra's Algorithm
B. Floyd-Warshall Algorithm
C. Prim's Algorithm
D. Bellman-Ford Algorithm

In the Bellman-Ford algorithm, how many times are the edges relaxed?

- A. $V - 1$ B. E C. V D. $E - 1$

Which of the following algorithms can find the shortest paths between all pairs of vertices in a graph?

- A. Bellman-Ford Algorithm
B. Dijkstra's Algorithm
C. Floyd-Warshall Algorithm
D. Kruskal's Algorithm

What is the time complexity of the Floyd-Warshall Algorithm?

- A. $O(V^2)$ B. $O(V^3)$ C. $O(E \log V)$ D. $O(V \log V)$

Which of the following data structures is used in Dijkstra's Algorithm to store vertices?

- A. Queue B. Stack
C. Priority Queue D. Deque

What is the key property of a greedy algorithm?

- A. Solve all subproblems optimally
B. Always makes the locally optimal choice
C. Divides the problem into smaller subproblems
D. Repeatedly solves the same subproblem

Which algorithm is used for finding the minimum spanning tree of a graph?

- A. Floyd-Warshall Algorithm
B. Bellman-Ford Algorithm
C. Prim's Algorithm
D. BFS

Which algorithm solves the all-pairs shortest path problem in a weighted directed graph?

- A. Prim's Algorithm
B. Kruskal's Algorithm
C. Floyd-Warshall Algorithm
D. BFS

What is the time complexity of DFS for a graph represented as an adjacency list?

- A. $O(V^2)$ B. $O(V + E)$
C. $O(E \log V)$ D. $O(V \log V)$

Which of the following techniques is used in solving the Longest Increasing Subsequence problem?

- A. Greedy B. Divide and Conquer
C. Dynamic Programming D. Brute Force

In which case does Kruskal's algorithm perform better than Prim's algorithm?

- A. When the graph has more edges
B. When the graph is sparse
C. When the graph is dense
D. When the graph is directed

What is the time complexity of Dijkstra's Algorithm for a graph represented with an adjacency matrix?

- A. $O(V^2)$ B. $O(E \log V)$
C. $O(V \log V)$ D. $O(V)$

Which of the following is not a valid sorting algorithm?

- A. Bubble Sort
B. Merge Sort
C. Binary Sort
D. Quick Sort

Which algorithm is commonly used in finding the minimum spanning tree for a graph?

- A. Dijkstra's Algorithm
B. Floyd-Warshall Algorithm
C. Kruskal's Algorithm
D. BFS

Which algorithm is used for matrix exponentiation efficiently?

- A. Divide and Conquer
B. Greedy Algorithm
C. Dynamic Programming
D. Brute Force

Which of the following can detect cycles in a graph?

- A. BFS B. DFS C. Kruskal's Dijkstra's

What is the time complexity of finding the Longest Common Subsequence (LCS) of two sequences of lengths m and n ?

- A. $O(m + n)$ B. $O(m * n)$
C. $O(m \log n)$ D. $O(n \log m)$

Which sorting algorithm uses the concept of divide and conquer?

- A. Bubble Sort B. Insertion Sort
C. Quick Sort D. Selection Sort

In which type of graph is Dijkstra's algorithm not guaranteed to work correctly?

- A. Graphs with negative weight edges
B. Directed Acyclic Graphs
C. Complete Graphs
D. Undirected Graphs

In Prim's algorithm, what is the time complexity if the graph is represented with an adjacency list and a priority queue is used?

- A. $O(V^2)$
B. $O(E \log V)$
C. $O(V \log V)$
D. $O(E)$

In the 0/1 Knapsack problem, which of the following is true?

- A. Items can be divided
B. Items cannot be divided
C. Only fractional items are allowed
D. Weights are ignored

Which algorithm is based on dynamic programming and is used for the shortest path with negative weights?

- A. Bellman-Ford Algorithm
B. Dijkstra's Algorithm
C. Prim's Algorithm
D. Kruskal's Algorithm

Which algorithm finds all-pairs shortest paths in $O(V^3)$?

- A. Prim's Algorithm
B. Bellman-Ford Algorithm
C. Kruskal's Algorithm
D. Floyd-Warshall Algorithm

Which problem can be solved by matrix exponentiation?

- A. Fibonacci sequence
B. Shortest Path Problem
C. Minimum Spanning Tree
D. 0/1 Knapsack

In Kruskal's algorithm, edges are added to the minimum spanning tree based on which criteria?

- A. Weight
B. Degree
C. Vertex number
D. Level of hierarchy

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Numerical Methods Laboratory

Course Code: CSE 206, Lab Final Examination

Scenario 1:

You are working in a civil engineering firm tasked with calculating the depth at which a pile must be driven into the ground. The formula that models the soil resistance $f(x)=x^3-4x-9$ is given, where x is the depth (in meters). You must find the depth where resistance equals zero.

Task:

- Use the **Bisection Method** to find the root of $f(x)=x^3-4x-9$ in the interval $[2,3]$
- Tolerance: $\epsilon=0.001$
- Maximum Iterations: 25
- Show all iterations in a table (Excel or code output)

Scenario 2:

You are a software engineer working on a climate model, and you need to solve a nonlinear equation involving temperature prediction:

$$f(x)=\ln(x)+x^2-3$$

Task:

- Use the Secant Method to find the root of the function
- Initial guesses: $x_0=0.5$, $x_1=1.0$
- Tolerance: $\epsilon=0.001$
- Limit iterations to 15
- Tabulate all steps: x_0 , x_1 , x_2 , $f(x_2)$, and error

Scenario 3

You're developing a software system for trajectory prediction of a drone. You are given the drone's height at different times:

$t=[1,2,4]$ seconds

$h(t)=[2.0,3.5,1.0]$ meters

Task:

- Use Lagrange's interpolation to find the height of the drone at $t=3$ seconds.
- Show all Lagrange basis polynomials and steps.
- Implement the solution in Excel or write a program.

Scenario 4:

As a civil engineer, you're estimating the area under a velocity-time curve to calculate the total distance traveled by a vehicle. The data is given below:

Time (s)	Velocity (m/s)
0	0
2	6
4	12
6	14
8	10
10	5

Task:

- Use the Trapezoidal Rule to estimate the total distance traveled in 10 seconds.
- Clearly show formula and each trapezoid calculation.